



Adventure Works Data Report

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INTRODUCTION

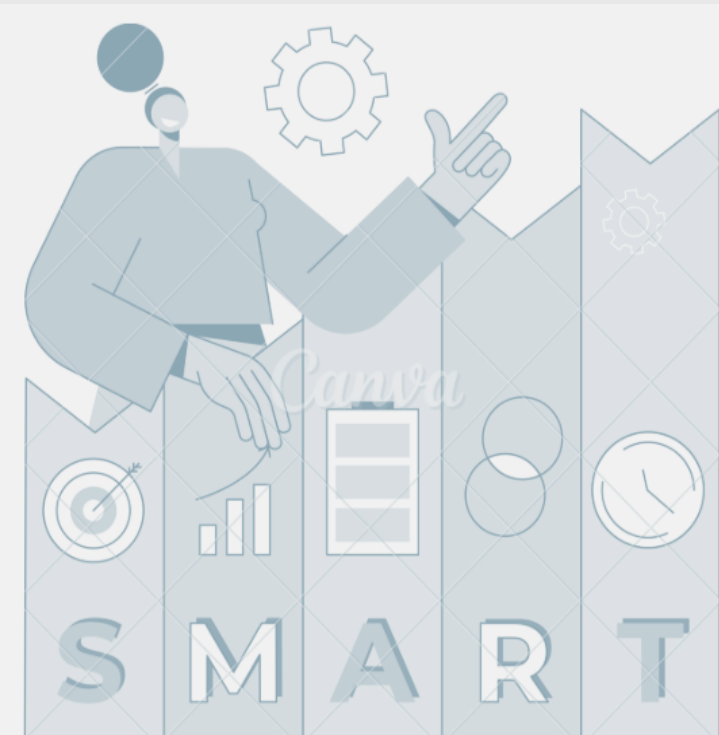


The Adventure Works Data Analytics Project focuses on analyzing sales performance, customer behavior, and product trends through an end-to-end analytics workflow.

Project Goals:

- Perform data cleaning, transformation, and preparation.
- Build relational data models and analytical structures.
- Analyze sales performance across yearly, quarterly, monthly, and financial periods.
- Examine weekday vs weekend behavior and seasonal trends.
- Evaluate regional and country-wise performance to identify high and low-performing markets.
- Identify both top-performing and underperforming products and subcategories.
- Study customer purchasing patterns using demographic analysis such as age groups, income distribution, gender, and occupation.

By transforming raw transactional data into structured analytical models and interactive dashboards, the project demonstrates how data analytics supports data-driven business decision-making.



DATA OVERVIEW



The data includes transactional sales records along with detailed product, customer, geographic, and time-related information.

Primary Fact Tables

FactInternetSales – Contains transactional sales data including order details, quantities, pricing, and revenue metrics.

FactInternetSalesNew – Additional sales records appended to the main fact table to create a complete sales dataset.

These tables form the core of the analysis and were combined during data preparation.



DATA OVERVIEW



Dimension Tables

DimCustomer – Customer demographic information such as gender, birthdate, yearly income, occupation, and geographic attributes.

DimProduct – Product-level details including product name, category references, and cost information.

DimProductSubCategory – Intermediate product hierarchy connecting products to categories.

DimProductCategory – High-level product classification for category analysis.

DimSalesTerritory – Geographic information including country and region for regional performance analysis.

DimDate – Calendar and time-related data used for time-series analysis and financial period calculations.



OBJECTIVES & KPIs



Project Objectives

- Analyze overall sales performance using integrated data from multiple datasets.
- Study sales trends across different time periods (year, month, weekday vs weekend).
- Identify both **top-performing and underperforming products** to evaluate product performance.
- Examine customer purchasing behaviour through customer analysis metrics.
- Transform raw transactional data using Excel, SQL, Tableau and Power BI into meaningful business insights.
- Build interactive dashboards to support data-driven decision making.

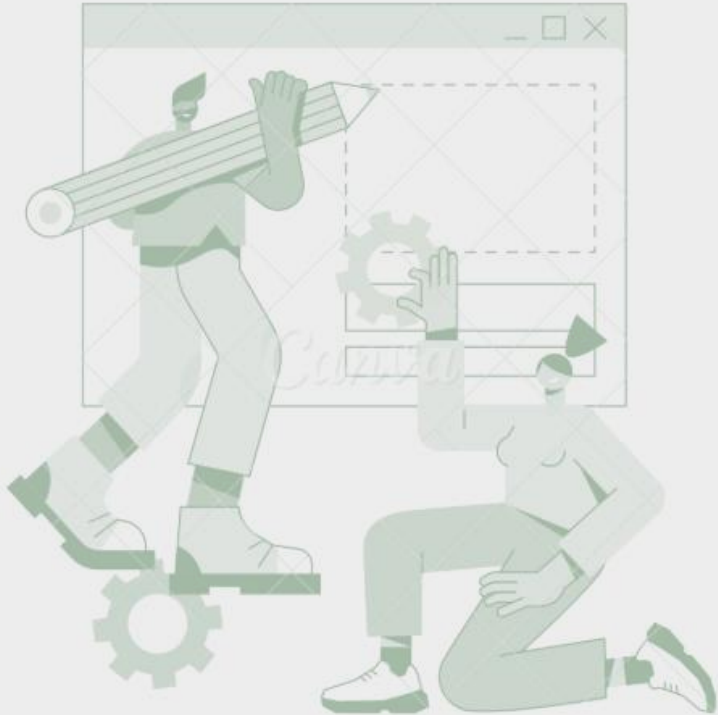


OBJECTIVES & KPIs



Key Performance Indicators (KPIs)

- Total Sales Revenue
- Total Profit
- Total Quantity Sold
- Average Sales per Customer
- Sales Trend Over Time (Monthly/Yearly)
- Weekday vs Weekend Sales Performance
- Top vs Bottom Product Performance
- Customer Purchase Contribution

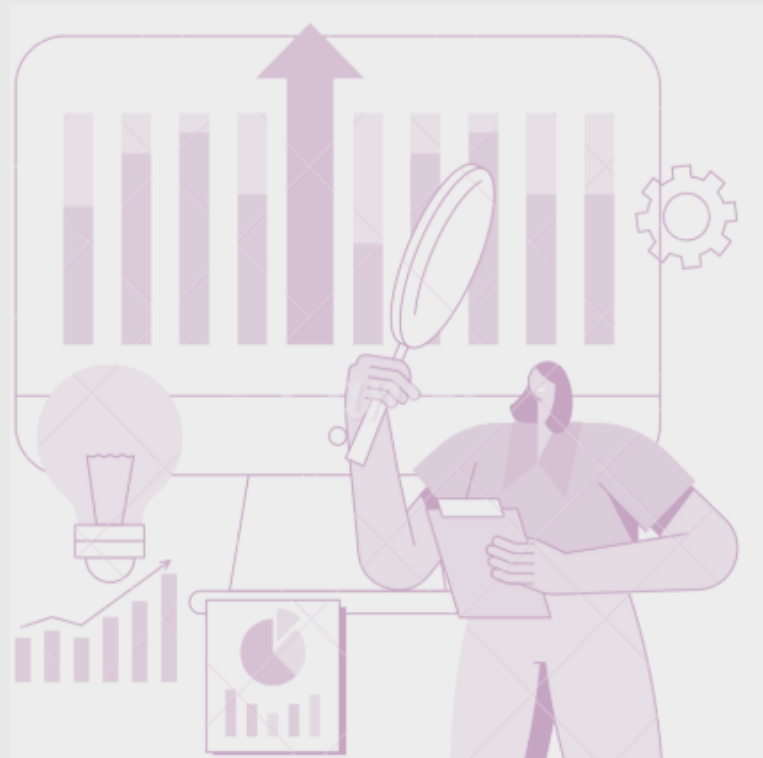


PROCESS



Data Cleaning ,Modelling & Preperation

- Imported multiple datasets related to sales, products, customers, and dates.
- Reviewed dataset structure, columns, and relationships before analysis.
- Removed unwanted columns and handled missing values.
- Corrected data types (dates, numeric fields).
- Established structured relationships between fact and dimension tables using a Snowflake schema with hierarchical product dimensions
- Created additional fields such as:
 - Customer age based on first purchase date
 - Age groups
 - Income categories
 - Weekday vs weekend classification



PROCESS

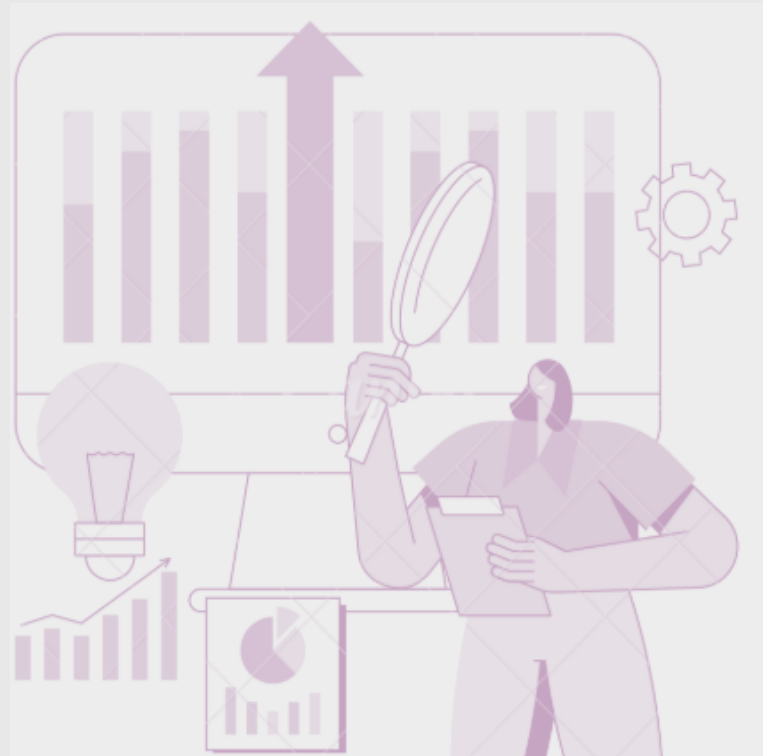


KPI Development

- Defined key performance indicators for business analysis.
- Calculated sales performance, order behaviour, and customer metrics.
- Evaluated both top-performing and bottom-performing products.

Analysis & Visualization

- Built analytical dashboards to explore trends, patterns, and customer behaviour.
- Derived meaningful insights to support data-driven decision making.



TREND ANALYSIS

Yearly Sales Trend

Sales grew strongly from 2010 to 2013, peaking in 2013, followed by a noticeable decline indicating possible market or operational shifts.



Month-wise Sales Trend

Sales increase gradually across the year with mid-year and year-end peaks, showing clear seasonal patterns.



Quarter-wise Sales Distribution

Q1 contributes the highest sales share, while Q4 performs relatively lower, suggesting improvement opportunities.



Day-wise Sales (Weekday vs Weekend)

Sales are higher on weekdays, indicating business-driven purchasing, while weekends show potential for growth strategies.



Top Regions & Country-wise Sales



Top Regions by Sales

- Australia leads with **9.1M** sales, followed by Southwest (**5.7M**) and Northwest (**3.6M**).
- United Kingdom (**3.4M**) and Germany (**2.9M**) show moderate performance.

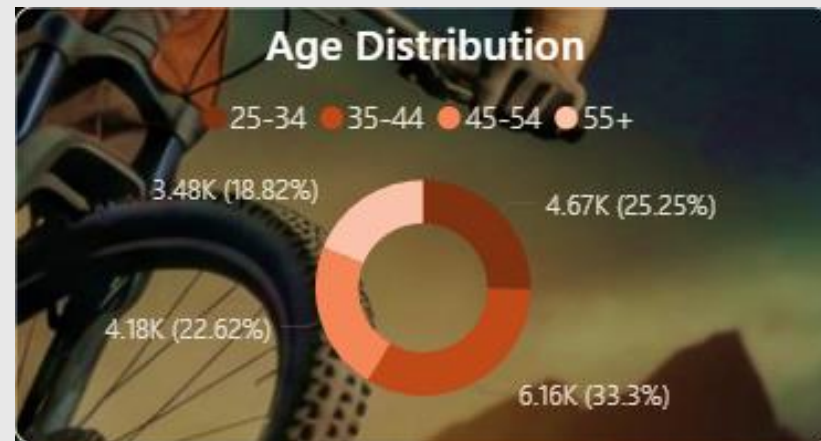
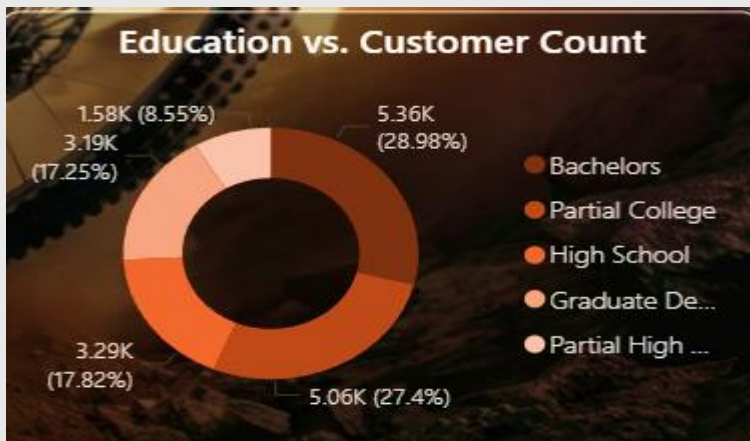
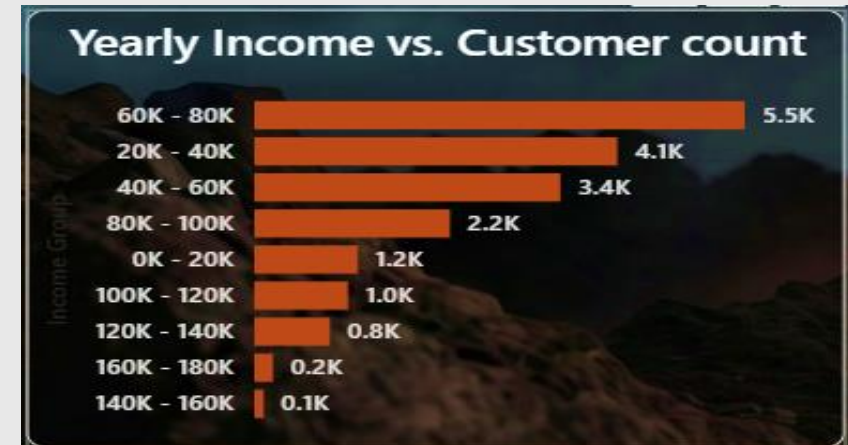
Country-wise Sales

- United States is the top market with **9.39M** sales.
- Australia follows closely with **9.06M**.
- UK (**3.39M**), Germany (**2.89M**), and France (**2.64M**) show steady contribution.
- Canada records the lowest sales at **1.98M**, indicating growth opportunity.



Customer Distribution

- Professional and Skilled Manual occupations represent the largest customer segments, indicating strong purchasing power among working professionals.
- Majority of customers fall within the mid-income ranges (20K–80K), showing core target customer groups.
- Age distribution is balanced, with the 25–44 age group contributing the highest share.
- Gender distribution is nearly equal, indicating balanced engagement across male and female customers.
- Customers with Bachelor's and Partial College education form a major portion of the customer base.



Top performing products & Top Sub categories

- Mountain and Road bike models dominate top-performing products, contributing the highest sales and profit.
- High-value bike products drive overall revenue compared to accessories.
- Road Bikes generate the highest sales among subcategories, followed by Mountain Bikes.
- Touring Bikes contribute moderate revenue, while accessories like Helmets and Tires show lower sales contribution.
- Sales performance is highly concentrated in core bike categories.



Top 10 Products		
ProductName	Total Sales	Total Profit
Mountain-200 Black, 46	1.37M	0.63M
Mountain-200 Black, 42	1.36M	0.62M
Mountain-200 Silver, 38	1.34M	0.61M
Mountain-200 Silver, 46	1.30M	0.59M
Mountain-200 Black, 38	1.29M	0.59M
Mountain-200 Silver, 42	1.26M	0.57M
Road-150 Red, 48	1.21M	0.47M
Road-150 Red, 62	1.20M	0.47M
Road-150 Red, 52	1.08M	0.42M
Road-150 Red, 56	1.06M	0.42M

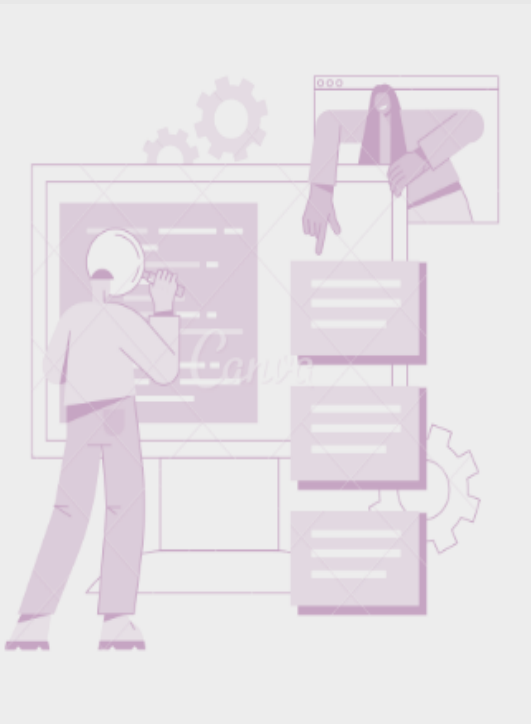


INSIGHTS & RECOMMENDATIONS



Key Insights

- Sales show strong growth from **2010 to 2013**, peaking in **2013**, with clear seasonal variation across months and quarters (Q1 highest contribution).
- Sales activity is higher during weekdays compared to weekends, indicating workday-driven purchasing behavior.
- Revenue is concentrated in key markets, with **United States (9.39M)** and **Australia (9.06M)** leading total sales.
- Core bike categories — especially **Road Bikes and Mountain Bikes** — contribute the largest share of revenue and profit.
- Customer distribution is strongest among **mid-income groups (20K–80K)** and **25–44 age group**, with balanced gender participation.
- Certain products and accessories show consistently lower performance, indicating opportunities for optimization or repositioning.

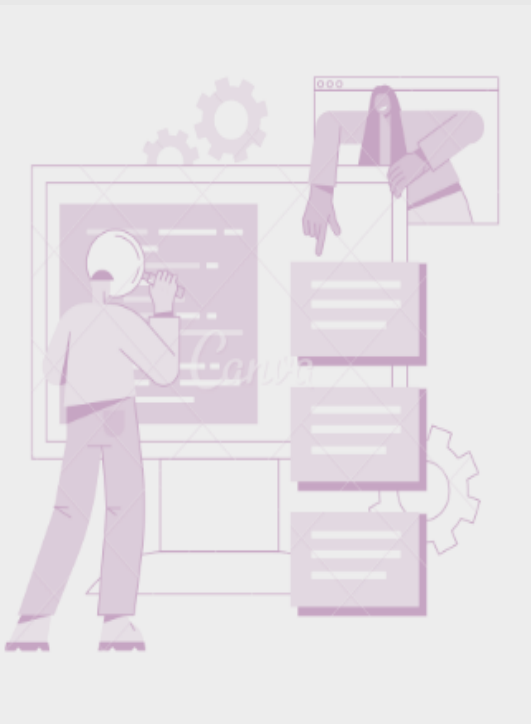


INSIGHTS & RECOMMENDATIONS



Recommendations

- Focus marketing and inventory planning around peak periods, as sales grew significantly from **2010–2013**, with peak performance in **2013**.
- Strengthen strategies in high-performing markets like the **United States (9.39M)** and **Australia (9.06M)** while exploring growth opportunities in lower-performing regions such as **Canada (1.98M)**.
- Increase promotion and availability of top-performing bike categories (especially **Road Bikes and Mountain Bikes**) which contribute the majority of revenue.
- Review pricing or promotional strategies for lower-performing products and accessories to improve contribution.
- Use demographic insights — strong participation from **mid-income groups (20K–80K)** and **25–44 age group** — to design targeted marketing campaigns.



CONCLUSION



The analysis of the sales dataset provides valuable insights into sales performance, customer behavior, product performance, and regional trends across different time periods.

- Sales show clear **seasonal and time-based patterns**, with identifiable yearly, monthly, and quarterly trends.
- Certain regions and countries consistently contribute higher revenue, highlighting **geographical performance differences**.
- Customer demographics such as age, income, occupation, education, and gender reveal distinct purchasing patterns.
- Both **top-performing and underperforming products and subcategories** were identified, supporting strategic product evaluation.

Overall, this project demonstrates how structured data modeling and analytical exploration can transform raw business data into actionable insights that support data-driven decision-making.



Challenges & Learnings

Challenges

- Cleaning and preparing raw data for analysis (data types, missing fields, formatting).
- Designing correct table relationships and understanding the schema structure.
- Choosing between calculated columns and aggregated measures for KPIs.
- Creating time-based trends with proper sorting and hierarchy handling.
- Building derived segments like age groups and income ranges from raw data.

Key Learnings

- Strong understanding of data modelling and relational structures.
- Improved analytical thinking and business-focused interpretation.
- Practical experience in feature engineering and segmentation.
- Better visualization design for clear insight communication.





Thank you